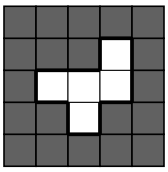
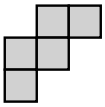
Preview: set-03/by_level/Grade_03-04/y2003_q18 (v3)

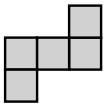
Covering a shape with pieces of squares: count the squares, then turn the pieces

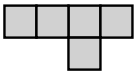
Part A: Each question is worth 3 points

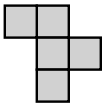
1. Which piece covers the empty area exactly? The pieces may be turned and flipped over.

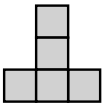


A. 

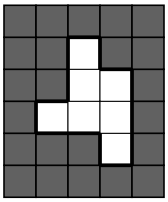
B. 

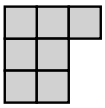
C. 

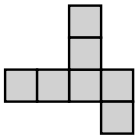
D. 

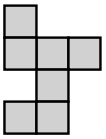
E. 

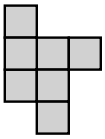
2. Which piece covers the empty area exactly? The pieces may be turned and flipped over.

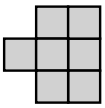


A. 

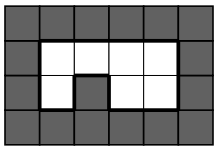
B. 

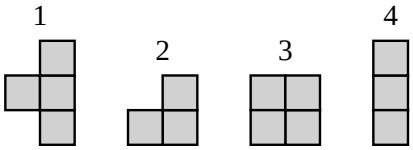
C. 

D. 

E. 

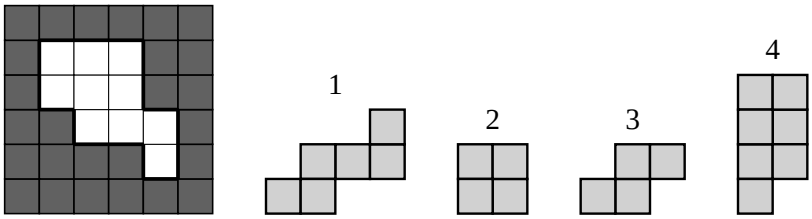
3. Which two of these pieces can be used to cover the empty area exactly? The pieces may be turned and flipped over.





A. 1+2 ***B.*** 1+4 ***C.*** 2+3 ***D.*** 2+4 ***E.*** 3+4

4. Which two of these pieces can be used to cover the empty area exactly? The pieces may be turned and flipped over.

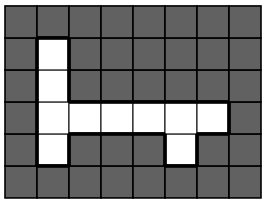


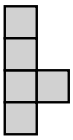
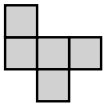
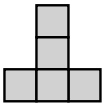
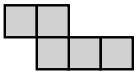
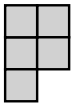
A. 1+2 ***B.*** 1+3 ***C.*** 1+4 ***D.*** 2+3 ***E.*** 2+4

Part B: Each question is worth 4 points

5. Two pieces that are exactly the same cover the empty area.

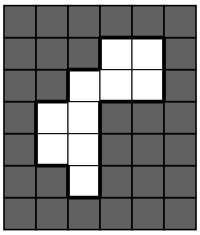
Which piece is it? The pieces may be turned and flipped over.

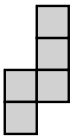
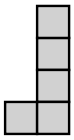
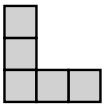
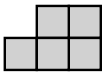
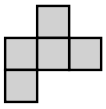


A.  ***B.***  ***C.***  ***D.***  ***E.*** 

6. Two pieces that are exactly the same cover the empty area.

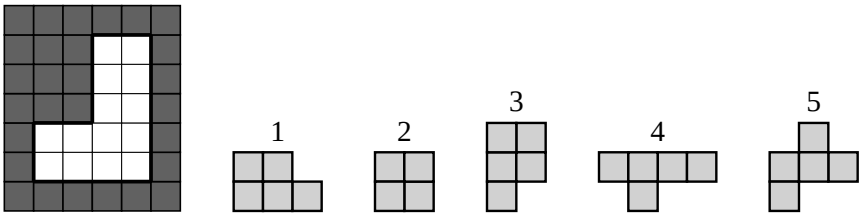
Which piece is it? The pieces may be turned and flipped over.



A.  ***B.***  ***C.***  ***D.***  ***E.*** 

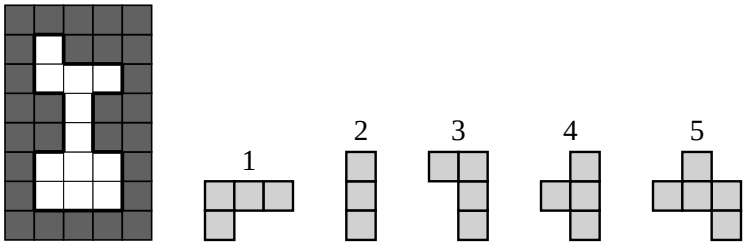
Part C: Each question is worth 5 points

7. Which three of these pieces can be used to cover the empty area exactly? The pieces may be turned and flipped over.



A. 1+2+3 ***B.*** 1+3+5 ***C.*** 1+4+5 ***D.*** 2+3+4 ***E.*** 2+3+5

8. Which three of these pieces can be used to cover the empty area exactly? The pieces may be turned and flipped over.



A. 1+2+3 ***B.*** 1+3+4 ***C.*** 1+3+5 ***D.*** 2+3+5 ***E.*** 2+4+5

Answer Key

1. **D**	2. **D**	3. **C**	4. **B**	5. **A**	6. **D**
7. **A**	8. **E**				

Solutions

1. D — The empty area has 5 squares; all pieces do too, so look at the shape. Turn piece D and it fits exactly; each of the others has one square in a different place. [set-03/by_level/Grade_03-04/2003 Q18 · one_piece]

2. D — The empty area has 7 squares; all pieces do too, so look at the shape. Turn piece D and it fits exactly; each of the others has one square in a different place. [set-03/by_level/Grade_03-04/2003 Q18 · one_piece]

3. C (2+3) — The empty area has 7 squares. Counting squares: piece 1 has 4, piece 2 has 3, piece 3 has 4, piece 4 has 3. Only pieces 2 and 3 also fit together in the right shape. [set-03/by_level/Grade_03-04/2003 Q18 · two_pieces]

4. B (1+3) — The empty area has 10 squares. Counting squares: piece 1 has 6, piece 2 has 4, piece 3 has 4, piece 4 has 7. Only pieces 1 and 3 also fit together in the right shape. [set-03/by_level/Grade_03-04/2003 Q18 · two_pieces]

5. A — The empty area has 10 squares, so each piece has 5; every option has 5. Only two copies of piece A fit together into the shape of the hole. [set-03/by_level/Grade_03-04/2003 Q18 · two_copies]

6. D — The empty area has 10 squares, so each piece has 5; every option has 5. Only two copies of piece D fit together into the shape of the hole. [set-03/by_level/Grade_03-04/2003 Q18 · two_copies]

7. A (1+2+3) — The empty area has 14 squares, so the three sizes must add up to 14. Only 1 + 2 + 3 also fit together in the right shape. [set-03/by_level/Grade_03-04/2003 Q18 · three_pieces]

8. E (2+4+5) — The empty area has 12 squares, so the three sizes must add up to 12. Only 2 + 4 + 5 also fit together in the right shape. [set-03/by_level/Grade_03-04/2003 Q18 · three_pieces]